Gynecologic Oncology 183 (2024) 1–6

Contents lists available at ScienceDirect

Gynecologic Oncology

journal homepage: www.elsevier.com/locate/ygyno

Integration and usability of a digital cancer risk stratification tool to optimize identification of patients at risk for hereditary cancers: A pilot study

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HIGHLIGHTS

- A digital cancer risk stratification tool assessed genetic testing eligibility and lifetime breast cancer risk.
- Uptake of the digital cancer risk stratification tool was high and usability scores were favorable.
- Age, health literacy, primary insurance, and race and ethnicity were associated with perceived tool usability.

ARTICLE INFO

Article history:
Received 20 December 2023
Received in revised form 19 February 2024
Accepted 22 February 2024
Available online 8 March 2024

Keywords:
Risk assessment
Genetics
HEREDITARY CANCER
Software

ABSTRACT

Background. Patients with a personal or family history of cancer may have elevated risk of developing future cancers, which often remains unrecognized due to lapses in screening. This pilot study assessed the usability and clinical outcomes of a cancer risk stratification tool in a gynecologic oncology clinic.

Methods. New gynecologic oncology patients were prompted to complete a commercially developed personal and family history-based risk stratification tool to assess eligibility for genetic testing using National Comprehensive Cancer Network criteria and estimated lifetime breast cancer risk using the Tyrer-Cuzick model. After use of the risk stratification tool, usability was assessed via completion rate and the System Usability Scale, and health literacy was assessed using the BRIEF Health Literacy Screening Tool.

Results. 130 patients were prompted to complete the risk stratification tool; 93 (72%) completed the tool. Race and ethnicity and insurance type were not associated with tool completion. The median System Usability Scale score was 83 out of 100 (interquartile range, 60–95). Health literacy positively correlated with perceived usability. Public insurance and race or ethnicity other than non-Hispanic White was associated with lower perceived usability. Sixty (65%) patients met eligibility criteria for genetic testing, and 21 (38% of 56 eligible patients) were candidates for enhanced breast cancer screening based on an estimated lifetime breast cancer risk of ≥20%.

Conclusions. A majority of patients completed the digital cancer risk stratification tool. Older age, lower health literacy, public insurance, and race or ethnicity other than non-Hispanic White were associated with lower perceived tool usability.

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1. Introduction

Over 1% of individuals in the United States carry a cancer-predisposing germline pathogenic variant, responsible for hereditary cancer syndromes such as Hereditary Breast and Ovarian Cancer Syndrome, Lynch Syndrome, and Li Fraumeni Syndrome [1]. Identification of individuals who are at increased risk for cancer or who harbor a cancer-predisposing pathogenic variant allows for the utilization of risk reducing methods. Such methods include cancer screening, which may diagnose cancer at earlier stages, and surgery or chemoprevention, which decreases the likelihood of developing cancer. These measures have been demonstrated to reduce cancer-related morbidity and mortality [2–7].

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https://doi.org/10.1016/j.ygyno.2024.02.028
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Multiple national organizations, including the National Comprehensive Cancer Network (NCCN®), suggest eligibility criteria for patients for genetic testing based on personal or family history [8–19]. In addition, irrespective of genetic status and identification of pathogenic variants, individuals with strong family history of cancer may be at elevated risk for cancer themselves and may warrant increased cancer screening. NCCN® recommends enhanced breast cancer screening using mammogram and breast MRI for individuals with a ≥20% estimated lifetime risk of breast cancer based on validated models such as Tyrer-Cuzick, Gail, and BRCAPro [20–24].

Despite known interventions for patients at elevated risk of cancer, many patients are not adequately screened for cancer risk. Among patients with cancer-predisposing pathogenic variants, evidence suggests as few as 5% have been identified [25]. Similarly, adequate family histories are taken in as few as one third of patients, limiting the ability to make accurate assessments of cancer risk [26–28]. In addition to insufficient collection of personal and family history, lack of adequate screening is also attributed to limited provider knowledge and appointment time constraints, which may hinder the identification of patients at increased risk [29]. Lack of adequate screening and testing exists not only in the primary care setting but among patients with gynecologic cancers, with less than half of patients with ovarian cancer receiving adequate genetic services [28,30].

Patient-facing tools have been developed to aid in the collection of relevant history and estimation of cancer risk [31–37]. These tools may overcome barriers to cancer risk assessment by lessening clinician burden regarding collection and interpretation of personal and family history. While several such tools have been integrated into practice, there exists limited data exploring patient-perceived usability and clinical impact of such tools with regard to cancer risk assessment. The purpose of this pilot study was to assess the usability and clinical outcomes of cancer risk stratification tool in a gynecologic oncology clinic.

2. Methods

2.1. Overview

This was a prospective quality improvement intervention at Weill Cornell Medicine to assess the usability of a digital genetic cancer risk stratification tool prior to potential scaling of the tool to other clinics and specialties. Retrospective review of patient characteristics and outcomes between November 2022 and May 2023 was approved by the Weill Cornell Medicine Institutional Review Board. Consecutive new patients presenting to gynecologic oncology appointments were prompted to complete a risk stratification tool. The gynecologic oncology practice focused on cancer genetics and cancer risk reduction. Prompts to complete the tool were provided either via email or text prior to the appointment, or at the time of arrival to the appointment via QR code presented by an office navigator. When the tool was completed, providers received a summary report with notification of genetic testing eligibility and estimated lifetime breast cancer risk with supporting clinical data. The summary was available to the clinician via the electronic medical record or, if preferred, via a secure online provider portal.

2.2. Digital cancer risk stratification tool

The piloted risk stratification tool was the Ambry Genetics Comprehensive, Assessment, Risk, and Education (CARE) Program™ [38]. CARE is a digital, patient-facing tool which combines artificial intelligence-driven Tyrer-Cuzick model (version 8.0) with complex, rules-guided, flow logic based on patient responses to pre-programmed input options designed to collect personal and family medical history. The digital tool is compatible with computers, smartphones, tablets, and other electronic devices. Responses are integrated to assess for eligibility for National Comprehensive Cancer Network (NCCN®) Clinical Practice Guidelines in Oncology genetic testing criteria for hereditary cancer syndromes associated with breast, ovarian, pancreatic, and colorectal cancers, as well as lifetime risk for breast cancer based on the Tyrer-Cuzick (version 8.0) [20–23,39,40]. Patients were required to enter their name and date of birth and create an account to proceed with the risk assessment (Supplemental Fig. 1). At completion of the risk assessments, patients had the option of completing educational video modules through CARE™ on relevant topics such as cancer risk and genetic testing at the conclusion of the risk assessment (Supplemental Fig. 2). During the study period, the tool was available in English and Spanish.

CARE was selected as the digital risk stratification tool by the quality improvement team as few available tools provided both an assessment of genetic testing eligibility for an array of hereditary cancer syndromes using NCCN Clinical Practice Guidelines in Oncology as well a lifetime breast cancer risk assessment. The adherence to NCCN Clinical Practice Guidelines in Oncology and concurrent assessment of lifetime breast cancer risk was in line with the practice's standard of care. Prior tools have been evaluated by the study group that lacked the potential for streamlined comprehensive assessment available through CARE [36,41].

2.3. Outcomes

The primary objective of this study was to understand the usability of the risk stratification tool. The completion rate was determined by the proportion of patients receiving tool-generated risk assessment out of the total number of patients prompted to complete the tool. Patients completing the tool were also requested to complete the System Usability Scale regarding their experience using the tool. The System Usability Scale is a widely used validated 10-item survey with high reliability, used to measure perceived usability of a tool [42–44]. Statements were graded on a scale of 1 (strongly disagree) to 5 (strongly agree) to generate a raw score between 1 and 100. The score was converted to a corresponding letter grade (A+ to F) for ease of interpretation as per the standard System Usability Scale scoring system. If an item was missing, a neutral score would be substituted for the missing items as per standard System Usability Scale practice [42].

The secondary objective was to understand factors associated with tool usability and downstream clinical outcomes after tool use. Patient characteristics were evaluated for associations with tool usability. Health literacy was measured using the BRIEF Health Literacy Screening Tool, which was offered to patients who completed the risk stratification tool [45]. The BRIEF Health Literacy Screening tool is a validated, 4-item measure which grades health literacy on a scale of 4 (lowest health literacy) to 20 (highest health literacy). Other patient characteristics were obtained via chart review. Downstream clinical outcomes included counseling regarding genetic testing eligibility for patients meeting NCCN® genetic testing criteria and counseling regarding enhanced breast cancer screening for those with estimated lifetime breast cancer ≥20% [20,21,39,40].

2.4. Statistical analysis

To explore the association between tool usability and patient characteristics, correlation between continuous variables was measured using Kendall's rank correlation test. Association between continuous and categorical variables was measured using the two-sample Wilcoxon rank-sum test. Pearson's Chi-squared test or Fisher's exact test, as appropriate for category size, were used to evaluate associations between categorical variables. Statistical significance was evaluated at the 0.05 alpha level. Statistical analysis was performed using R software (Version 3.6.1 [07/05/19], R Foundation for Statistical Computing, Vienna, Austria).

3. Results

3.1. Patient characteristics

Between November 2022 and May 2023, 130 new gynecologic oncology patients were prompted to complete the risk stratification tool (Fig. 1). The median age of the patients was 50 years (interquartile range [IQR] 39–64 years; Table 1). The patients were primarily female (129, 99%) and identified as non-Hispanic White (95, 73%), Asian (13, 10%), non-Hispanic Black (9, 6.9%), Hispanic (8, 6.2%), or other (5, 3.9%). Regarding primary insurance, 74 (57.0%) patients had private or commercial insurance, 28 (22%) had Medicare or managed Medicare, 15 (12%) had Medicaid or managed Medicaid, 12 (9.2%) had other state-funded insurances, and 1 (0.8%) was self-pay. English was largely the preferred language (119, 92%). Most patients received a prompt to complete the tool via email or text message prior to arrival at the office (95, 73%), while a minority accessed the tool via a QR code presented by an office navigator upon arrival to the office (35, 27%). Sixty-six (51%) patients were referred to the practice specifically for genetic and risk-reducing services (referrals included known cancer-predisposing pathogenic variant, family history of cancer, and family history of cancer-predisposing pathogenic variant), while 64 (49%) patients were referred for reasons unrelated to available genetic services (including new diagnosis of gynecologic cancer, gynecologic precancer, and other gynecologic complaints).

3.2. Risk stratification tool completion

The tool was completed by 93 patients (72%). Among completers, 46 (49%) proceeded to complete the available educational modules. A lower median age was associated with tool completion (47 versus 53 years, $p = 0.024$; Supplemental Fig. 3). Those prompted via text message or email to complete the tool prior to the day of appointment were more likely to complete the tool than those prompted to complete the tool via QR code at the office (80% versus 57%, $p = 0.008$). Patients with a family history of cancer were more likely to complete the tool (95% versus 76%, $p = 0.003$). Among the 37 that did not complete the tool, the reasons for lack of completion were technical difficulty (10, 27%), lack of interest (8, 22%), lack of computer/smartphone access (5, 14%), and language barrier (2, 5.4%); 12 patients (32%) did not specify reason for lack of completion. No difference was found between rate of completers versus non-completers for race and ethnicity, primary insurance type, and preferred language.

3.3. System Usability Scale scores

Among those who completed the risk stratification tool, 88 (94.6%) completed the System Usability Scale (Table 2). The median System Usability Scale score was 83 (IQR 60–95), which corresponded to a letter grade of A (IQR D-A+). Age was inversely correlated with System Usability Scale score (Kendall's $\tau = -0.178$, $p = 0.018$; Supplemental Fig. 4 A). Among the 88 patients who completed the BRIEF Health Literacy Screening tool, the median score was 18.5 (IQR 16–20). Health literacy positively correlated with the System Usability Scale score (Kendall's $\tau = 0.210$, $p = 0.010$; Supplemental Fig. 4B). Individuals with private insurance had higher System Usability Scale scores than those without (median score 88 [letter grade of A+] versus 75 [letter grade of B], $p = 0.039$; Supplemental Fig. 5 A), and patients identifying as non-Hispanic White had higher System Usability Scale scores than those who did not (71 [letter grade of C] versus 89 [letter grade of A+], $p = 0.008$; Supplemental Fig. 5B). The median System Usability Scale score did not differ by preferred language ($p = 0.673$).

3.4. Genetic testing

The risk stratification tool identified 60 (65%) patients who met eligibility criteria for genetic testing based on personal or family history, of whom 42 (70%) had previously undergone testing (Fig. 1). Of the remaining 18 patients eligible for testing, all received counseling regarding germline genetic testing, and 16 (89%) opted to undergo multi-gene panel genetic testing on the day of the appointment. Genetic testing identified 3 (19%) patients with cancer-predisposing pathogenic variants (*CHEK2*, *FH*, *ATM*), and 1 (6.3%) patient with a variant of uncertain significance in a cancer-predisposing gene (*AXIN2*).

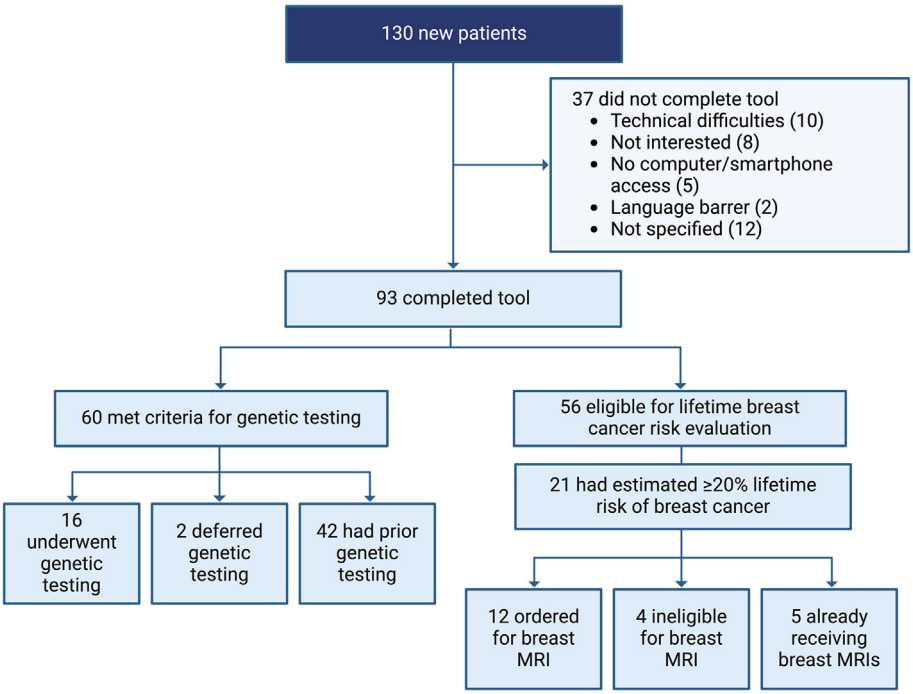


Fig. 1. Study flowchart. Created with BioRender.com.

Table 1
Patient characteristics and tool completion. IQR; interquartile ratio. *$n = 88$.

Patient characteristic	Total ($n = 130$)		Completers ($n = 93$)		Non-Completers ($n = 37$)		
	Median	IQR	Median	IQR	Median	IQR	*p*-value
Age in years	50	39–64	47	38–62	53	44–69	**0.024**
System Usability Scale score*	–	–	83	60–95	–	–	–
BRIEF Health Literacy Scale score*	–	–	18.5	16–20	–	–	–
	n	%	n	%	n	%	
Sex							
Female	129	99	92	99	37	100	>0.999
Male	1	0.8	1	1.1	0	0	
Race and ethnicity							
Non-Hispanic White	95	73	68	73	27	73	
Asian	13	10	11	12	2	5.4	
Non-Hispanic Black	9	6.9	7	7.5	2	5.4	0.408
Hispanic	8	6.2	5	5.4	3	8.1	
Other	5	3.9	2	2.2	3	8.1	
Primary insurance							
Private insurance	74	57	57	61	17	46	
Medicare/managed Medicare	28	22	19	20	9	24	
Medicaid/managed Medicaid	15	12	8	8.6	7	19	0.167
Other state-funded insurance	12	9.2	9	9.7	3	8.1	
Self-pay	1	0.8	0	0.0	1	2.7	
Preferred language							
English	119	92	88	95	31	84	0.075
Other	11	8.5	5	5.4	6	16	
Tool prompt							
Email/text message	95	73	74	80	21	57	**0.008**
QR code	35	27	19	20	16	43	
Personal history of cancer							
No	79	61	60	65	19	51	0.165
Yes	51	39	33	35	18	49	
Family history of cancer							
No	14	11	5	5	9	24	**0.003**
Yes	116	89	88	95	28	76	
Referral for genetic services							
No	64	49	43	46	21	43	0.279
Yes	66	51	50	54	16	57	

3.5. Breast cancer screening

Among tool completers, 56 (60%) were eligible to have estimated lifetime breast cancer risk calculations performed via the Tyrer-Cuzick model (Fig. 1). Of eligible patients, 21 (38%) met criteria for enhanced breast cancer screening with estimated lifetime breast cancer risk of ≥20%. Twelve (57%) patients were recommended enhanced breast cancer screening and were ordered for breast MRI, 5 (24%) were already receiving enhanced breast cancer screening, 3 (14%) were counseled but too young to initiate enhanced breast cancer screening, and 1 (4.8%) had already undergone risk-reducing bilateral mastectomy.

4. Conclusions

In this pilot quality improvement study, a commercially developed, digital pre-appointment, patient-facing risk stratification tool was integrated into gynecologic oncology office practice for evaluation of new patients. The tool was generally perceived as usable by patients with a completion rate of over 70% and a median usability score that corresponded to an "A" letter grade. While older age, identification as race or ethnicity other than non-Hispanic White, public insurance, and lower health literacy were associated with decreased perceived tool usability, decreased likelihood of tool completion was only associated with older age and lack of family cancer history. Nearly two-thirds of patients met NCCN® criteria for genetic testing, and 60% had a lifetime breast cancer risk of ≥20%. All patients who met criteria for genetic testing or enhanced breast cancer screening were counseled by their clinician regarding recommended interventions.

Multiple barriers exist to completing a risk assessment for inherited germline cancer pre-disposing pathogenic variants and lifetime risk of breast cancer. Such barriers often revolve around the collection of relevant personal and family history, which is time-consuming but a critical cornerstone of cancer risk estimation. Favorable outcomes have been demonstrated with the use of tools to collect personal and family history prior to scheduled appointments, with evidence suggesting that tool-based history collection may lead to higher rates of guideline-appropriate recommendations being made by clinicians [36]. However, with complex algorithms and guidelines dictating which patients meet eligibility criteria, collection of adequate history does not necessarily lead to patients receiving appropriate counseling and recommendations [29,36,46]. The risk stratification tool in this study serves as both a tool to collect relevant personal and family history, as well as a means of calculating and communicating cancer risk. Recognition of those with above-average cancer risks is critical to enable patients to pursue means of cancer risk reduction and prevention which can reduce morbidity and mortality.

CARE™, the risk stratification tool used in this study, was previously described in Loving et al., with use among women undergoing breast imaging [32]. Loving et al., reported on downstream clinical outcomes of the use of CARE™, including genetic testing and uptake of enhanced breast cancer screening. However, the authors did not offer data regarding response rate, patient perception regarding use of the tool, or factors associated with tool usability. In our study, the reporting of completion rates and validated usability scores allows for measures of both objective and subjective usability. Older age was associated with a decreased likelihood of completing the tool, and lower usability scores correlated with older age, public insurances, and identification as a race and ethnicity other than non-Hispanic White. Such trends are consistent with characteristics that have been known to be associated with decreased access to genetic services [47–53].

Table 2
System Usability Scale score by patient characteristic. IQR; interquartile ratio. $N = 88$.

Patient characteristic	System Usability Scale score		
	Median	IQR	p-value
Sex			
Female ($n = 88$)	83	60–95	-
Male ($n = 0$)	-	-	
Race and ethnicity			
Non-Hispanic White ($n = 64$)	89	68–98	**0.008**
Not non-Hispanic White ($n = 24$)	71	49–88	
Primary insurance			
Private insurance ($n = 53$)	88	70–98	**0.039**
Public insurance ($n = 35$)	75	52–90	
Preferred language			
English ($n = 84$)	85	59–96	0.673
Other ($n = 4$)	80	75–80	
Tool prompt			
Email/text message ($n = 69$)	85	60–95	0.647
QR code ($n = 19$)	80	59–93	
Personal history of cancer			
No ($n = 57$)	88	70–98	0.061
Yes ($n = 31$)	70	48–95	
Family history of cancer			
No ($n = 5$)	78	50–100	0.971
Yes ($n = 83$)	83	60–95	
Referral for genetic services			
No ($n = 41$)	77	55–93	0.156
Yes ($n = 47$)	88	68–98	
Criteria for genetic testing met			
No ($n = 30$)	86	58–92	0.951
Yes ($n = 58$)	79	62–98	
Estimated lifetime risk of breast cancer			
<20% ($n = 32$)	80	59–98	0.628
≥20% ($n = 21$)	80	68–92	

While clinical significance of such correlations remains unclear, care must be taken to ensure the implementation of a digital risk assessment tool does not facilitate further disparity. However, prior literature suggests that the use of digital genetic cancer risk assessment tools may facilitate screening among populations that have historically been underserved with regard to genetic services [41]. Further, exploration of the accuracy of the history collected by the risk assessment tool by the patient compared to the history obtained via clinician interview may be warranted, particularly among populations that report lower perceived tool usability. Our institution is conducting a randomized controlled trial evaluating same risk stratification tool within an urban benign gynecology clinic (NCT05562778), which is anticipated to enroll a population that has historically had decreased accessed to genetic services. This ongoing study may provide insight into the role of digital risk stratification tools in addressing disparity within genetic cancer risk assessment.

Limitations of this study include the observational design. This study took place in a genetics-focused gynecologic oncology office where patients may be uniquely motivated to share cancer-related personal and family history. Although patients who had a family history of cancer were more likely to complete the risk stratification tool, no differences were seen in completion rate or usability score between patients referred specifically for genetic services and those referred for non-genetic gynecologic oncologic indications. This lack of difference supports the generalizability of the risk stratification tool to a broader gynecologic oncology population. However, the uptake of genetic testing and enhanced cancer screening may be elevated compared to a general gynecologic population given the large degree of referrals for genetic services. Additionally, the lack of statistical significance identified during the exploratory analyses of the associations between tool completion rates and patient demographics should be interpreted with caution, given the relatively small numbers in some cohorts and the retrospective analysis of the data. Lastly, the study sought to identify associations between patient characteristics and tool usability but did not aim to identify specific barriers faced by patients that may be at the root cause of lower completion rates or perceived usability. Qualitative research may be warranted to gain a more nuanced understanding of such issues.

Our study demonstrates favorable usability for a digital health tool, and similar risk stratification tools hold promise to expand the accessibility of cancer risk assessment in the outpatient setting. Further research on the use of risk stratification tools is being conducted in a randomized controlled manner (NCT05562778) and is anticipated to provide valuable information regarding the impact, effectiveness, and accessibility of risk stratification tools in a diverse general population.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ygyno.2024.02.028.

Disclosures

Melissa K Frey was supported by the following grant: American Association of Obstetricians and Gynecologists Foundation / American Board of Obstetrics & Gynecology (AAOGF/ABOG) Career Development Award. Ravi Sharaf was supported by the following grants: National Cancer Institute Grant # K07CA216326 and R01CA211723 and Patient Centered Outcomes Research Institute Grant # IHS-2017C3–9211. The funding sources had no role in study design; collection, analysis, and interpretation of data; in the writing of the report; or in the decision to submit the article for publication.

Presentation information

A subset of this data was presented as a poster at the 2023 Annual ASCO Meeting, Chicago, IL on June 3, 2023.

CRediT authorship contribution statement

Emily M. Webster: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luiza Perez:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Muhammad Danyal Ahsan:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sarah Levi:** Writing – review & editing, Software, Data curation, Conceptualization. **Isabelle Chandler:** Writing – review & editing, Visualization, Validation, Supervision, Project administration. **Charlene Thomas:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Kemi Babagbemi:** Writing – review & editing, Methodology, Conceptualization. **Ravi N. Sharaf:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology. **Melissa K. Frey:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors report no conflicts of interest.

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